

According to Niels Birbaumer, a neuroscientist who led the research at University of Tübingen, Germany, "It is the first sign that completely locked-in syndrome may be



The cap uses infrared light to spot variations in blood flow in different regions of the brain. A computer then learns to distinguish the patterns for yes and no for each patient (Image courtesy: [www.wysscencer.ch](http://www.wysscencer.ch))

abolished forever, because with all of these patients, we can now ask them the most critical questions in life."

During the sessions, patients wore a cap that uses infrared light to spot variations in blood flow in different regions of the brain. As they answered the questions, a computer hooked up to the cap learned to distinguish the blood flow patterns for yes and no in each patient.

## Robotic arm for minimally-invasive surgery

In an era of spiralling healthcare cost concerns, a new US\$ 500 surgical instrument developed at University of Michigan, USA, is vying to take the place of a US\$ 2 million robot for certain minimally-invasive procedures.

U-M startup FlexDex Surgical's first product, a simple, ergonomic and intuitive needle driver for stitching inside the body, has been used for the first time in a series of operations. FlexDex is an all-mechanical platform that mounts to the surgeon's arm. It uses a unique engineering approach to enable the tip of the instrument to mimic the direction of movement of the surgeon's hand.

Shorya Awtar, associate professor of mechanical engineering, says, "FlexDex provides the functionality of robots at the cost of traditional hand-held laparoscopic instruments. We have disrupted the paradigm where surgeons and hospitals had to choose between high cost/



FlexDex's platform technology precisely translates the surgeon's hand, wrist and arm movements from outside the patient into corresponding movements of an end-effector inside the patient's body (Image courtesy: <http://flexdex.com>)

high function and low cost/low function. Our mission is to democratise minimally-invasive surgery and expand its use around the USA and the world."

## Samsung TV that looks like framed art when turned off

At Consumer Electronics Show (CES) 2017 early this year, Samsung showcased its concept TV called The Frame, a TV that seamlessly blends into the background and displays



At CES 2017, Samsung showed off the concept for The Frame, a TV designed to double as a picture frame (Image courtesy: <http://www.businessinsider.in>)

artwork when not in use to further enhance living room décor. The Frame comes with more than 100 art pieces from ten categories, including landscape, architecture, wildlife, action, drawing and more. The TV also comes with a bunch of customisable accessories including interchangeable bezels and Studio Stand option, in case you do not wish to mount the TV on your wall.